Manual Verification

1) 8 Bit Addition : 8BH + 65H

8BH : 1000 1011

65H : 0110 0101

```
  1000   1011
+ 0110   0101
-----------------
  1111   0000
```

Result : F0H

CY = 0 Z = 0

AC = 1 S = 1

P = 1 OV = 0

Flag byte (OV, S, Z, AC, P, CY) : 1001 0100 = 94H

2) 8 Bit subtraction : 85H - 36H

85H : 1000 0101

36H : 0011 0110

```
  1000   0101
- 0011   0110
-----------------
  0100   1111
```

Result = 4FH

CY = 0 Z = 0

AC = 1 S = 0

P = 0 OV = 1

Flag bytes (OV, S, Z, AC, P, CY) = 0101 0000 = 50H

3) 16 Bit Addition : 2345H + 4567H

$$\begin{array}{r} 2345\,H \\ (+)\ 4567\,H \\ \hline 68ACH \end{array}$$

5 + 7 = C 3 + 5 = 8

4 + 6 = A 2 ÷ 4 = 6

Result : 68ACH, no carry beyond 16 bits

4) 16-Bit Subtraction : 2345H - 4567H

$$\begin{array}{r} 2345\,H \\ (-)\ 4567\,H \\ \hline -2222H \\ \hline \end{array}$$

1000 H - 2222 H = DDDE H

Result DDDC H, Cy = 1

5) 8 Bit Multiplication 14H x 06H

14 H = 20

06 H = 6

14 H x 06 H = 14 H + 14 H + 14 H + 14 H + 14 H + 14 H

= 78 H

Result = 78 H (20 x 6) = 120

6) 8 Bit Division : 48 H ÷ 06 H

48 H = 72

06 H = 6

72 ÷ 6 = 12

12 decimal = 0CH

Result

Quotient (Q) = 0CH

Remainder (R) = 00H

4) 16 - Bit multiplication : 2345H × 0003H

2345H × 3 : 2345H + 2345H + 2345H

```
  2345H
+ 2345H
+ 2345H
---------
  69CFH
```

Result = 0006 69CFH

5) 16 - Bit division : 4567H ÷ 1234H

4567H - 1234H = 3333H

3333H - 1234H = 20FFH

20FFH - 1234H = 0ECBH

0ECBH - 1231H < 0

Result :

Q = 0003H R = 0ECBH

3 × 1234H + 0ECBH = 4567H.